Algebraic quantum field theory

Hans Halvorson

November 12, 2013

Haag-Kastler axioms

Let $\mathcal{M}$ be Minkowski spacetime. For each double cone O, we assume that $\mathcal{A}(O) \subseteq \mathcal{A}$ is a C^*-algebra. The algebra $\mathcal{A}$ is called the **quasilocal** algebra of observables.

axiom 1 (isotony) If $O_1 \subseteq O_2$ then $\mathcal{A}(O_1) \subseteq \mathcal{A}(O_2)$.

axiom 2 (microcausality) If O_1 is spacelike separated from O_2 then $\mathcal{A}(O_1)$ commutes with $\mathcal{A}(O_2)$.

axiom 3 (translation covariance) There is a representation $x \mapsto \alpha_x$ of the translation group of $\mathcal{M}$ in the group of automorphism of $\mathcal{A}$ such that

$$\alpha_x(\mathcal{A}(O)) = \mathcal{A}(O + x).$$

Haag-Kastler axioms

positive energy vacuum representation: Let ω be a translation invariant state of $\mathcal{A}$, and let $(\mathcal{H}, \pi, x \mapsto U(x), \Omega)$ be the covariant GNS representation induced by ω. Let $\mathcal{R}(O) = \pi(\mathcal{A}(O))''$.

axiom 4 (additivity) For each double cone O, the set $\{\mathcal{R}(O + x) : x \in \mathbb{R}^4\}$ acts irreducibly on $\mathcal{H}$.

axiom 5 (spectrum condition) The four-momentum (P_0, P_1, P_2, P_3) has spectrum restricted to the forward light cone.

Theorem (Reeh-Schlieder) For any double cone O, the vacuum vector Ω is cyclic for $\mathcal{R}(O)$. That is, $\{A\Omega : A \in \mathcal{R}(O)\}$ is dense in the Hilbert space $\mathcal{H}$.

Proof uses an analyticity argument, invoking the spectrum condition.

consequences of Reeh-Schlieder

$\hookrightarrow$ the vacuum state is entangled; is highly entropic in local regions

$\hookrightarrow$ the vacuum vector is separating for localized operators: $A\Omega = 0$ only if $A = 0$

$\hookrightarrow$ no localized number operators

more consequences of the axioms

Theorem (Kadison) Local algebras are of infinite type.

$\hookrightarrow$ corollary: for any two spacelike separated regions, a dense set of states violates Bell's inequality

$\hookrightarrow$ all states are mixed (no normal states are pure)

$\hookrightarrow$ example: in the model of the free Boson field, local algebras are type III_1.

Bell's inequalities

Given C^*-algebras $\mathcal{A}$ and $\mathcal{B}$, such that $[A, B] = 0$ for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$.
A Bell operator is of the form

$$R = \frac{1}{2}(A_1(B_1 + B_2) + A_2(B_1 - B_2)),$$

where $A_i \in \mathcal{A}$ and $B_i \in \mathcal{B}$ are self-adjoint with $\|A_i\| \leq 1$ and $\|B_i\| \leq 1$.

Bell's theorem

The correlations for measurements of A_i and B_i in state ω permit a local hidden variable model just in case $|\omega(R)| \leq 1$.

Tsirelson bound: All quantum states satisfy $|\omega(R)| \leq \sqrt{2}$.

For $\mathcal{A} = M_2 \otimes I$ and $\mathcal{B} = I \otimes M_2$, and appropriate choice of spin operators A_i and B_i, the singlet state maximally violates Bell's inequality.

Nonlocality in QFT

Theorem (Summers and Werner) If $\mathcal{A}$ and $\mathcal{B}$ are the algebras associated with the Rindler wedges, then every state maximally violates Bell's inequality.

Theorem (Clifton and Halvorson) If $\mathcal{A}$ and $\mathcal{B}$ are algebras associated with any two spacelike separated regions, then the set of Bell-violating states is dense in the state space.